

RAG-Based Customer Support Assistant – High Level Design

1. System Overview

A RAG-based Customer Support Assistant that answers user queries using content from uploaded PDF documents by combining retrieval and LLM-based generation.

Scope:

- Upload PDFs
- Generate and store embeddings
- Retrieve relevant context
- Provide contextual answers
- Support escalation (HITL)

2. Architecture Components

- UI – Streamlit
- Document Ingestion – PDF Loader
- Chunking Module
- Embedding Model (HuggingFace)
- Vector Store – ChromaDB
- Retrieval System
- LLM Layer (Groq)
- LangGraph Workflow
- HITL Module

3. Component Description

- Loader – Extracts text from PDFs
- Chunking – Splits text into smaller units
- Embedding – Converts text to vectors
- Vector Store – Stores embeddings
- Retriever – Fetches relevant chunks
- LLM – Generates responses using context

- Workflow – Manages logic via LangGraph
- HITL – Enables human escalation

4. Data Flow

PDF → Chunking → Embedding → Storage → Query → Retrieval → LLM → Response

5. Technology Stack

- LangChain
- ChromaDB
- Streamlit
- HuggingFace
- Groq

6. Scalability

- Supports multiple documents
- Scalable vector storage
- Efficient semantic retrieval